Elements of Information Theory **2026-2027**

Probability measures, kernels and laws

Supplement to Lecture 1 *Lecturer: S. Yang*

This note is for the curious. It gives the measure-theoretic definitions behind the notation of Lecture 1: probability measures, random variables and their laws (distributions), densities, kernels (channels), and joint and output distributions. In particular, it explains the difference between a probability measure and a distribution: a distribution is always the distribution of a random variable, and it is defined through a probability measure. The lectures can be followed with pmfs and pdfs only; this note shows what the notation means in general.

1 Probability measures

A **measurable space** $(\mathcal{X}, \mathcal{A})$ is a set $\mathcal{X}$ with a σ-algebra $\mathcal{A}$ of subsets of $\mathcal{X}$, called **events**: $\mathcal{A}$ contains $\mathcal{X}$ and is closed under complement and countable union. On a finite or countable $\mathcal{X}$, $\mathcal{A}$ is the set of all subsets; on $\mathbb{R}^d$, it is the Borel σ-algebra, generated by the open sets.

A **measure** on $(\mathcal{X}, \mathcal{A})$ is a map $\mu\colon \mathcal{A} \to [0, \infty]$ with $\mu(\emptyset) = 0$ that is countably additive: $\mu\big(\bigcup_i A_i\big) = \sum_i \mu(A_i)$ for pairwise disjoint events $A_1, A_2, \ldots$. It is a **probability measure** if $\mu(\mathcal{X}) = 1$. Two measures that are not probability measures are used below: the **counting measure**, $\mu(A) = |A|$, on a countable $\mathcal{X}$, and the **Lebesgue measure** (length, area, volume) on $\mathbb{R}^d$.

$\mathcal{P}(\mathcal{X})$ is the set of probability measures on $(\mathcal{X}, \mathcal{A})$; the σ-algebra is left implicit. It is a convex set: for $\mathrm{P}_0, \mathrm{P}_1 \in \mathcal{P}(\mathcal{X})$ and $\lambda \in [0, 1]$, the mixture $(1-\lambda)\mathrm{P}_0 + \lambda \mathrm{P}_1$, defined event by event, is again a probability measure.

2 Random variables and their laws

Probability theory starts from an underlying **probability space** $(\Omega, \mathcal{F}, \mathbb{P})$: a set Ω of outcomes ω, a σ-algebra $\mathcal{F}$ of events, and a probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$. A **random variable** with values in $\mathcal{X}$ is a measurable map $\mathrm{X}\colon \Omega \to \mathcal{X}$, i.e., $\{\mathrm{X} \in A\} := \{\omega \in \Omega : \mathrm{X}(\omega) \in A\}$ is in $\mathcal{F}$ for every event A of $\mathcal{X}$.

The **law**, or **distribution**, of X is the probability measure on $\mathcal{X}$

$$\mathrm{P}_{\mathrm{X}}(A) := \mathbb{P}(\mathrm{X} \in A) = \mathbb{P}\big(\mathrm{X}^{-1}(A)\big), \qquad A \in \mathcal{A},$$

called the image of $\mathbb{P}$ under X. We write $\mathrm{X} \sim \mathrm{P}_{\mathrm{X}}$.

Probability measure versus distribution. A probability measure is defined on a measurable space, with no random variable involved: $\mathbb{P}$ on Ω, or any $\mathrm{P} \in \mathcal{P}(\mathcal{X})$. A distribution is always the distribution *of a random variable*: P_{X} is the probability measure on $\mathcal{X}$ that X carries over from $\mathbb{P}$. Every distribution is therefore a probability measure, defined through

$\mathbb{P}$ and X. Conversely, every $\mathrm{P} \in \mathcal{P}(\mathcal{X})$ is the distribution of some random variable: take $\Omega = \mathcal{X}$, $\mathbb{P} = \mathrm{P}$ and X the identity map, so that $\mathrm{P_X} = \mathrm{P}$ (the **canonical construction**). This is why $\mathcal{P}(\mathcal{X})$ is also called the set of distributions over $\mathcal{X}$, and why the lectures rarely mention Ω and $\mathbb{P}$.

Expectations depend only on the law. For a measurable $f\colon \mathcal{X} \to \mathbb{R}$, nonnegative or integrable,

$$\mathbb{E}\big[f(\mathrm{X})\big] = \int_\Omega f\big(\mathrm{X}(\omega)\big)\,\mathbb{P}(\mathrm{d}\omega) = \int_{\mathcal{X}} f(x)\,\mathrm{P_X}(\mathrm{d}x).$$

For an indicator $f = \mathbf{1}_A$, this is the definition of $\mathrm{P_X}$; the general case follows by linearity and monotone limits. This is why the lecture writes $\mathbb{E}_{\mathrm{P_X}}\big[f(\mathrm{X})\big]$, or $\mathbb{E}_{\mathrm{P}}$, with the distribution only. In particular, the information measures of Lecture 1 are functions of distributions, e.g., $H(\mathrm{X}) \equiv H(\mathrm{P_X})$.

One random variable, several distributions. The same map X has another law under another probability measure on $(\Omega, \mathcal{F})$: under $\mathbb{P}$ and $\mathbb{Q}$, its laws are $\mathrm{P_X}(A) = \mathbb{P}(\mathrm{X} \in A)$ and $\mathrm{Q_X}(A) = \mathbb{Q}(\mathrm{X} \in A)$. With the canonical construction ($\Omega = \mathcal{X}$, X the identity), $\mathrm{P_X} = \mathbb{P}$ and $\mathrm{Q_X} = \mathbb{Q}$: choosing a distribution in $\mathcal{P}(\mathcal{X})$ is choosing the probability measure under which X is considered. In the lecture, $\mathrm{X} \sim \mathrm{Q_X}$ or $\mathbb{E}_{\mathrm{Q}}[\,\cdot\,]$ says which one is used. For instance, $D(\mathrm{P}\|\mathrm{Q}) = \mathbb{E}_{\mathrm{P}}\big[\log \frac{p(\mathrm{X})}{q(\mathrm{X})}\big]$ is an expectation under P of a function of X built from both distributions.

Joint distributions. Two random variables X and Y on the same Ω form the random variable (X, Y) with values in $\mathcal{X} \times \mathcal{Y}$, whose events are generated by the rectangles $A \times B$. Its law $\mathrm{P_{XY}} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ is the **joint distribution**, and the **marginals** $\mathrm{P_X}(A) = \mathrm{P_{XY}}(A \times \mathcal{Y})$ and $\mathrm{P_Y}(B) = \mathrm{P_{XY}}(\mathcal{X} \times B)$ are the laws of X and Y. The random variables X and Y are **independent** if $\mathrm{P_{XY}} = \mathrm{P_X P_Y}$, the product measure, i.e., $\mathrm{P_{XY}}(A \times B) = \mathrm{P_X}(A)\,\mathrm{P_Y}(B)$ for all events A and B.

3 Densities

Let μ be a reference measure on $\mathcal{X}$, such as the counting measure or the Lebesgue measure. A distribution $\mathrm{P} \in \mathcal{P}(\mathcal{X})$ has the **density** $p \geq 0$ with respect to μ, written $p = \frac{\mathrm{dP}}{\mathrm{d}\mu}$, if

$$\mathrm{P}(A) = \int_A p\,\mathrm{d}\mu \quad \text{for every event } A, \qquad \text{and then} \qquad \mathbb{E}_{\mathrm{P}}\big[f(\mathrm{X})\big] = \int_{\mathcal{X}} f\,p\,\mathrm{d}\mu.$$

- Counting measure on a countable $\mathcal{X}$: $\int_A p\,\mathrm{d}\mu = \sum_{x \in A} p(x)$, and $p(x) = \mathrm{P}(\{x\})$ is the **pmf**.
- Lebesgue measure on $\mathbb{R}^d$: $\int_A p\,\mathrm{d}\mu = \int_A p(x)\,\mathrm{d}x$, and p is the **pdf**.

The lecture uses the same formulas in both cases, e.g., $\mathbb{E}_{\mathrm{P}}\big[\log \frac{1}{p(\mathrm{X})}\big]$ is $H(\mathrm{P})$ with the pmf and $h(\mathrm{P})$ with the pdf; only μ changes. A density is unique up to a μ-null set, which is why statements on pdfs hold "for almost every x".

Absolute continuity. P is **absolutely continuous** with respect to Q, written $\mathrm{P} \ll \mathrm{Q}$, if $\mathrm{Q}(A) = 0 \Rightarrow \mathrm{P}(A) = 0$ for every event A. If P and Q have densities p and q with respect to μ, then $\mathrm{P} \ll \mathrm{Q}$ iff $p = 0$ μ-almost everywhere on $\{q = 0\}$; for pmfs, this is $q(x) = 0 \Rightarrow p(x) = 0$ for every x, as in the lecture. Indeed, $\mathrm{Q}(\{q = 0\}) = 0$, so $\mathrm{P} \ll \mathrm{Q}$ gives $\int_{\{q=0\}} p \,\mathrm{d}\mu = 0$; conversely, if $\mathrm{Q}(A) = \int_A q \,\mathrm{d}\mu = 0$, then $q = 0$ μ-almost everywhere on A, hence so is p, and $\mathrm{P}(A) = 0$.

Radon–Nikodym derivative. By the Radon–Nikodym theorem, $\mathrm{P} \ll \mathrm{Q}$ iff P has a density with respect to Q, called the **Radon–Nikodym derivative** $\frac{\mathrm{dP}}{\mathrm{dQ}}$: $\mathrm{P}(A) = \int_A \frac{\mathrm{dP}}{\mathrm{dQ}} \,\mathrm{dQ}$ for every event A. If P and Q have densities p and q with respect to μ, then $\frac{\mathrm{dP}}{\mathrm{dQ}} = \frac{p}{q}$ on $\{q > 0\}$, a set of P-probability 1: this is the ratio p/q of the lecture, well defined P-almost surely. Two distributions always have densities with respect to a common measure, for instance $\mu = \mathrm{P} + \mathrm{Q}$.

Divergence and entropies. For $\mathrm{P} \ll \mathrm{Q}$, $D(\mathrm{P}\|\mathrm{Q}) = \mathbb{E}_{\mathrm{P}}\big[\log \frac{\mathrm{dP}}{\mathrm{dQ}}(\mathrm{X})\big] = \mathbb{E}_{\mathrm{P}}\big[\log \frac{p(\mathrm{X})}{q(\mathrm{X})}\big]$ for every μ that gives densities to P and Q: the divergence does not depend on the choice of μ, and it is defined for any two distributions, discrete, continuous or neither. The entropy and the differential entropy depend on the reference measure: $H(\mathrm{P})$ uses the pmf (counting measure) and $h(\mathrm{P})$ the pdf (Lebesgue measure). This is one reason why h behaves differently from H: it can be negative, and it changes with the scale, $h(a\mathrm{X}) = h(\mathrm{X}) + \log|a|$.

4 Kernels, joint and output distributions

A **(probability) kernel**, or **channel**, from $\mathcal{X}$ to $\mathcal{Y}$ is a family $\mathrm{P}_{\mathrm{Y}|\mathrm{X}} = (\mathrm{P}_{\mathrm{Y}|\mathrm{X}=x})_{x\in\mathcal{X}}$ of probability measures on $\mathcal{Y}$ such that $x \mapsto \mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}(B)$ is measurable for every event B of $\mathcal{Y}$; $\mathcal{P}(\mathcal{Y}|\mathcal{X})$ is the set of kernels from $\mathcal{X}$ to $\mathcal{Y}$. For finite alphabets, a kernel is a stochastic matrix: its row x is the pmf $p(\cdot \,|\, x)$ of $\mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}$, with nonnegative entries summing to 1. Like a probability measure, a kernel is defined without random variables; it becomes the conditional distribution of Y given X once (X, Y) is defined as below.

Joint distribution. For $\mathrm{P}_{\mathrm{X}} \in \mathcal{P}(\mathcal{X})$ and $\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \in \mathcal{P}(\mathcal{Y}|\mathcal{X})$, the joint distribution $\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ is the probability measure such that

$$(\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}})(A \times B) = \int_A \mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}(B) \,\mathrm{P}_{\mathrm{X}}(\mathrm{d}x) \qquad \text{for all events } A \text{ and } B;$$

these values determine it on all events of $\mathcal{X} \times \mathcal{Y}$. With densities, its density is $p(x)\, p(y \,|\, x)$. This defines a pair (X, Y) without choosing Ω: take $\Omega = \mathcal{X} \times \mathcal{Y}$, $\mathbb{P} = \mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$, and X and Y the two coordinates. Then $\mathrm{X} \sim \mathrm{P}_{\mathrm{X}}$, and $\mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}$ is the conditional distribution of Y given $\mathrm{X} = x$. This is how the lecture defines every distribution explicitly, in order: P_{X}, then $\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$, then $\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$.

Output distribution. The $\mathcal{Y}$-marginal of $\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$ is the output distribution

$$\begin{aligned}(\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{P}_{\mathrm{X}})(B) &:= (\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}})(\mathcal{X} \times B) \\ &= \int_{\mathcal{X}} \mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}(B) \,\mathrm{P}_{\mathrm{X}}(\mathrm{d}x) = \mathbb{E}_{\bar{\mathrm{X}}\sim\mathrm{P}_{\mathrm{X}}}\big[\mathrm{P}_{\mathrm{Y}|\mathrm{X}=\bar{\mathrm{X}}}(B)\big],\end{aligned}$$

a mixture of the distributions $\mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}$ with mixing distribution P_{X}. With pmfs, $p(y) = \sum_x p(x)\, p(y \,|\, x)$. Kernels compose in the same way: $\mathrm{P}_{\mathrm{Z}|\mathrm{Y}} \circ \mathrm{P}_{\mathrm{Y}|\mathrm{X}} := (\mathrm{P}_{\mathrm{Z}|\mathrm{Y}} \circ \mathrm{P}_{\mathrm{Y}|\mathrm{X}=x})_{x \in \mathcal{X}} \in \mathcal{P}(\mathcal{Z}|\mathcal{X})$ is the channel from $\mathcal{X}$ to $\mathcal{Z}$ through $\mathcal{Y}$; for finite alphabets, it is the product of the two stochastic matrices.

Conditional distributions and reverse channels. Conversely, a joint distribution $\mathrm{P}_{\mathrm{XY}} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ determines its marginals and, when $\mathcal{X}$ and $\mathcal{Y}$ are finite, countable or Euclidean spaces (more generally, standard Borel spaces), kernels such that

$$\mathrm{P}_{\mathrm{XY}} = \mathrm{P}_{\mathrm{X}} \mathrm{P}_{\mathrm{Y}|\mathrm{X}} = \mathrm{P}_{\mathrm{Y}} \mathrm{P}_{\mathrm{X}|\mathrm{Y}}$$

(**disintegration**). The conditional distributions $\mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}$ are unique only for P_{X}-almost every x: with pmfs, $p(y \,|\, x) = p(x, y)/p(x)$ when $p(x) > 0$, and $\mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}$ is arbitrary when $p(x) = 0$. This is why statements about conditional distributions hold P_{X}-almost surely. The kernel $\mathrm{P}_{\mathrm{X}|\mathrm{Y}}$ is the **reverse channel**; with pmfs, $p(x \,|\, y) = p(x)\, p(y \,|\, x)/p(y)$ (Bayes' rule).

Summary

Object	Definition	In the lecture
Probability measure	countably additive map from the events of $\mathcal{X}$ to $[0, 1]$, with total mass 1	$\mathrm{P}, \mathrm{Q} \in \mathcal{P}(\mathcal{X})$
Random variable	measurable map $\mathrm{X} \colon \Omega \to \mathcal{X}$ on $(\Omega, \mathcal{F}, \mathbb{P})$	$\mathrm{X}, \mathrm{Y}, \mathrm{Z}$
Law (distribution) of X	image of $\mathbb{P}$ under X: $\mathrm{P}_{\mathrm{X}}(A) = \mathbb{P}(\mathrm{X} \in A)$	P_{X}, $\mathrm{X} \sim \mathrm{P}_{\mathrm{X}}$
Density	$p = \mathrm{dP}/\mathrm{d}\mu$: the pmf for the counting measure, the pdf for the Lebesgue measure	$p(x)$, $q(x)$
Kernel (channel)	family $(\mathrm{P}_{\mathrm{Y}|\mathrm{X}=x})_{x \in \mathcal{X}}$ of probability measures on $\mathcal{Y}$, measurable in x	$\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \in \mathcal{P}(\mathcal{Y}|\mathcal{X})$
Joint distribution	$A \times B \mapsto \int_A \mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}(B)\, \mathrm{P}_{\mathrm{X}}(\mathrm{d}x)$	$\mathrm{P}_{\mathrm{X}} \mathrm{P}_{\mathrm{Y}|\mathrm{X}}$
Output distribution	$\mathcal{Y}$-marginal of the joint distribution, the mixture $\mathbb{E}_{\bar{\mathrm{X}} \sim \mathrm{P}_{\mathrm{X}}} \left[\mathrm{P}_{\mathrm{Y}|\mathrm{X}=\bar{\mathrm{X}}} \right]$	$\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{P}_{\mathrm{X}}$